Classical statistical inference: *confidence interval*

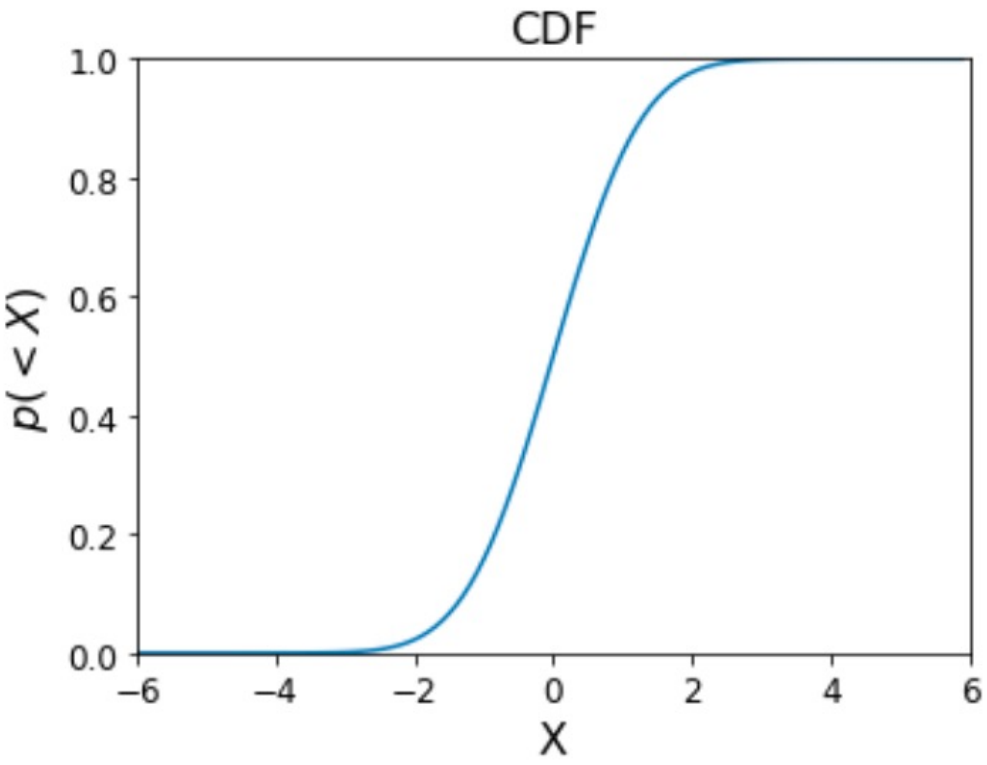
Part 3

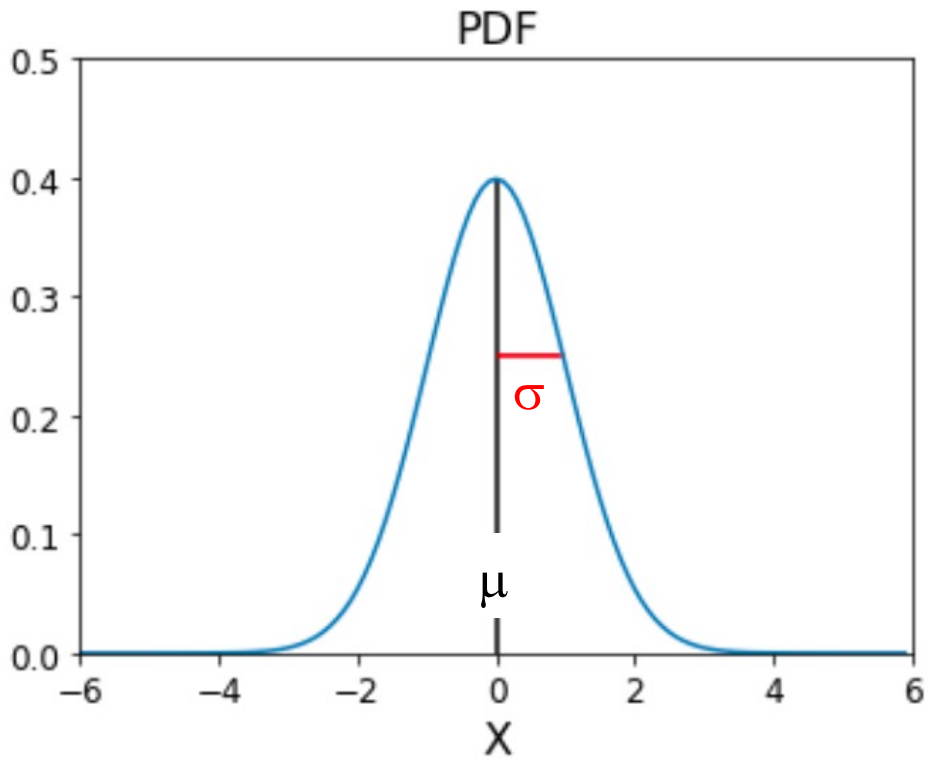
Associated notebook:

04-Basic_statistical_inference_frequentists_2/Frequentist_inference_02.ipynb

Confidence interval on a RV (descriptive stat)

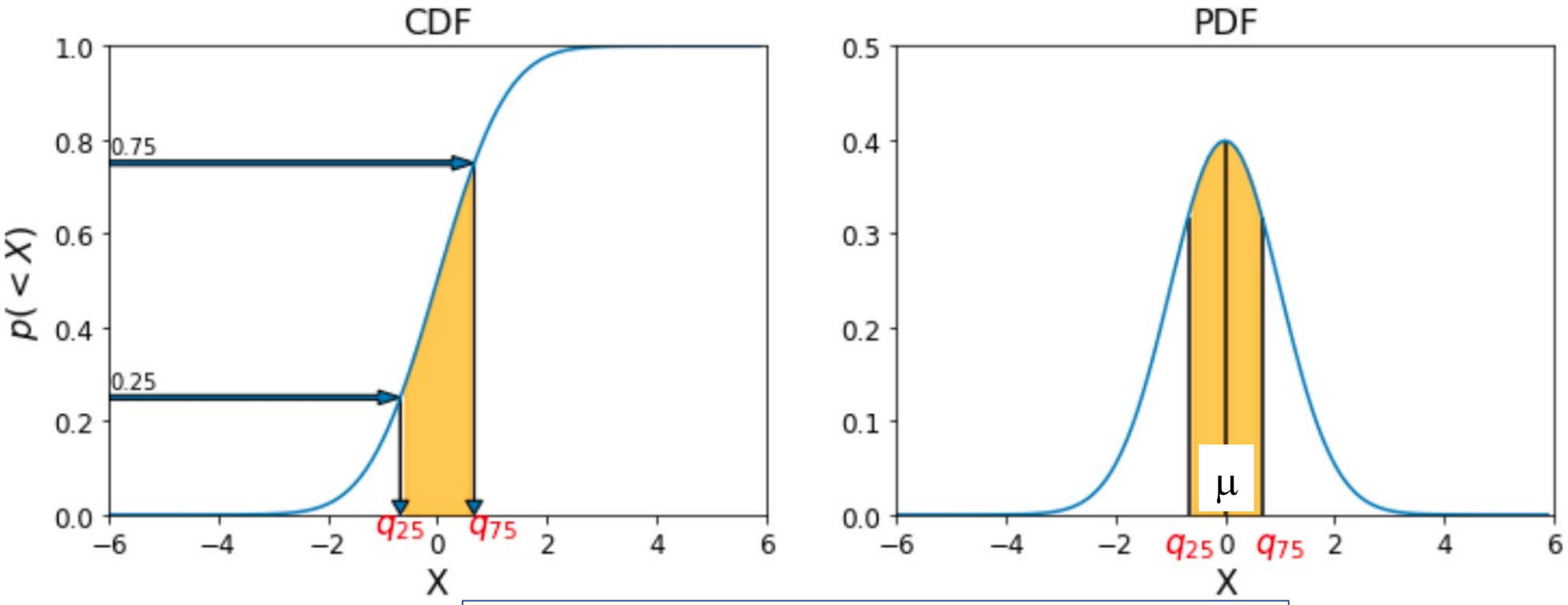
If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$





Confidence interval on a RV (descriptive stat)

If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$



Interquartile IQ = $[q_{25}, q_{75}] \equiv 50\ \%$ CI around μ

$1 - \alpha$ CI $\equiv [a, b]$ such that $p(a \leq X \leq b) = (1 - \alpha)$

Confidence interval on a RV (descriptive stat)

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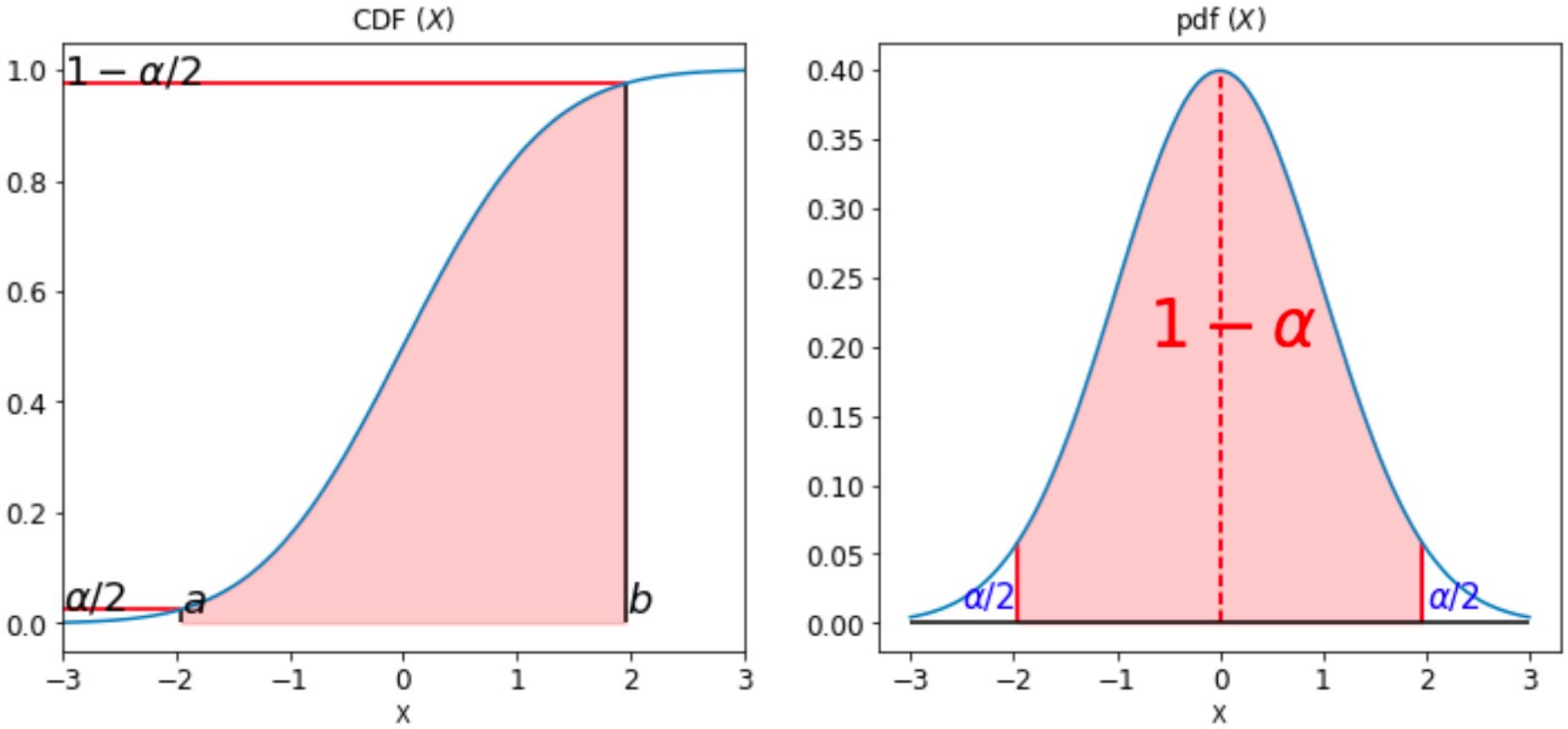

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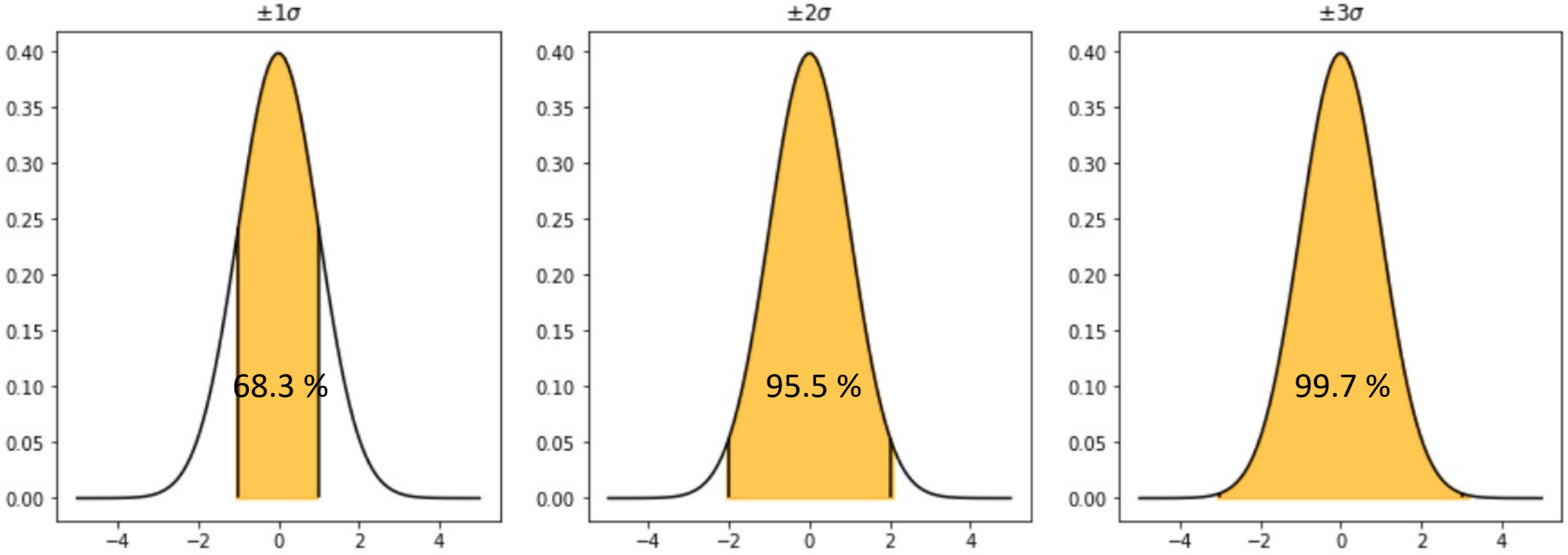


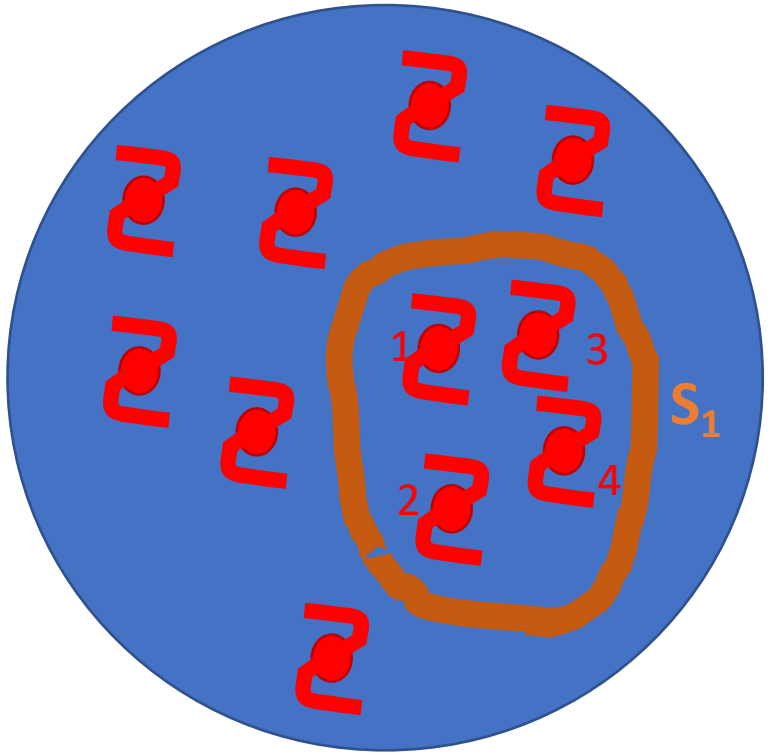
$[\mu - 1\ \sigma, \mu + 1\ \sigma] \equiv 68.3\ \%$

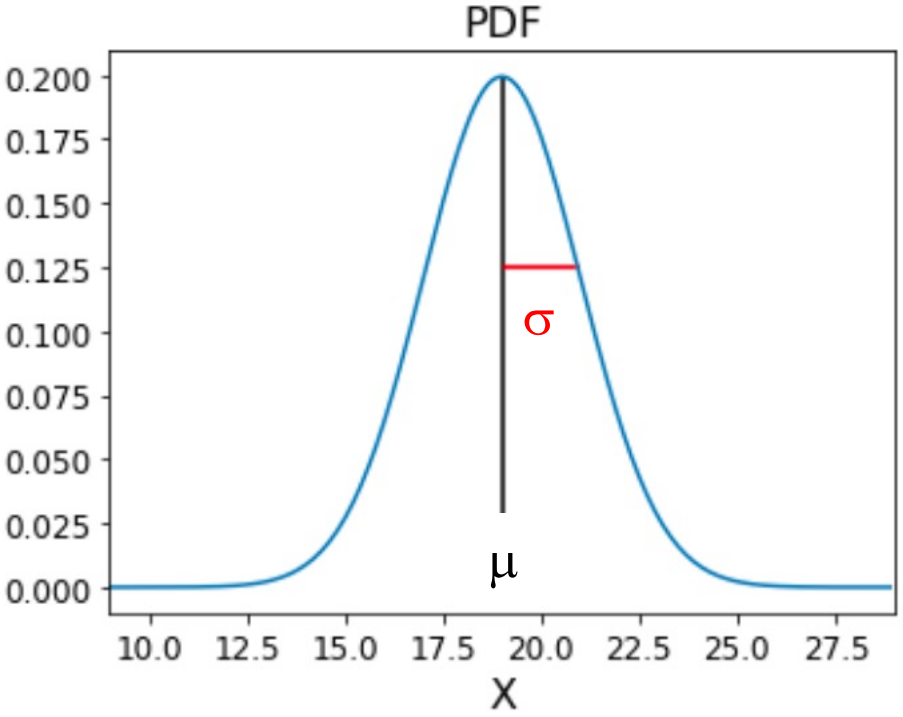
$[\mu - 2\ \sigma, \mu + 2\ \sigma] \equiv 95.5\ \%$

$[\mu - 3\ \sigma, \mu + 3\ \sigma] \equiv 99.7\ \%$

Confidence interval from an estimator

If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$

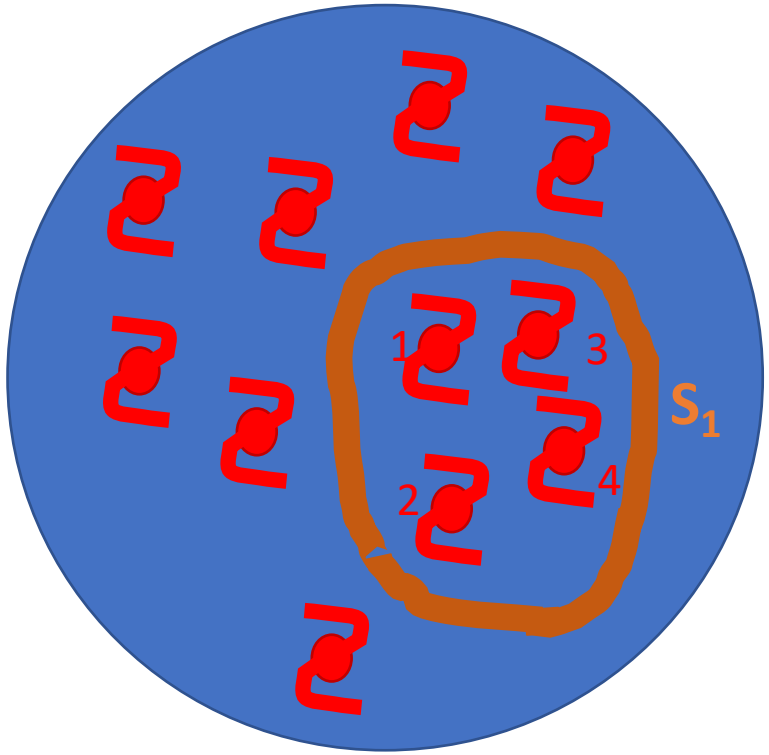


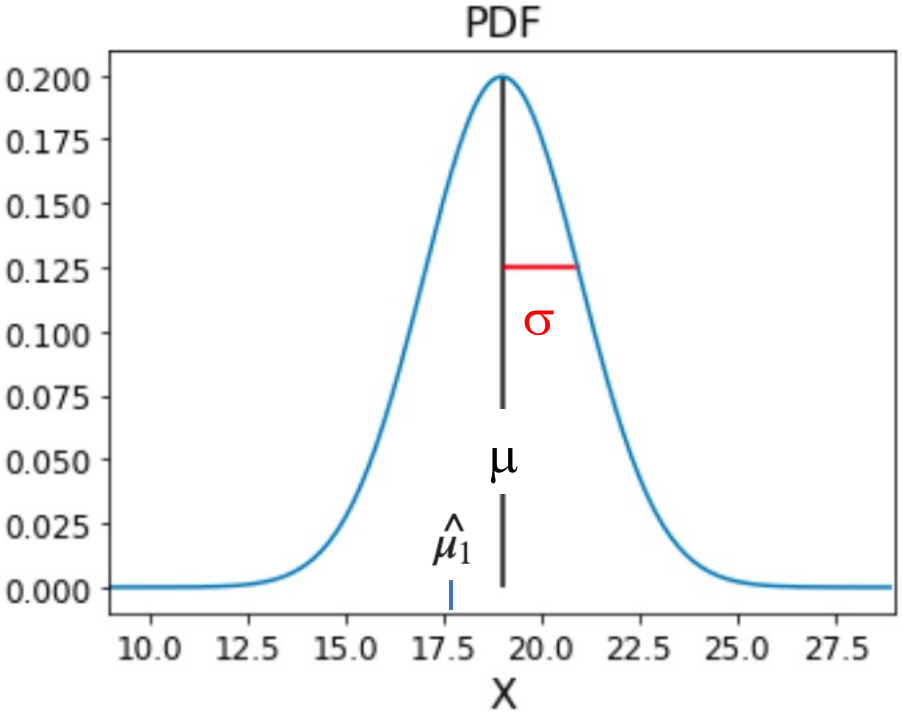


$S_1 = \{x_1, x_2, x_3, x_4\}$

Confidence interval from an estimator

If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$





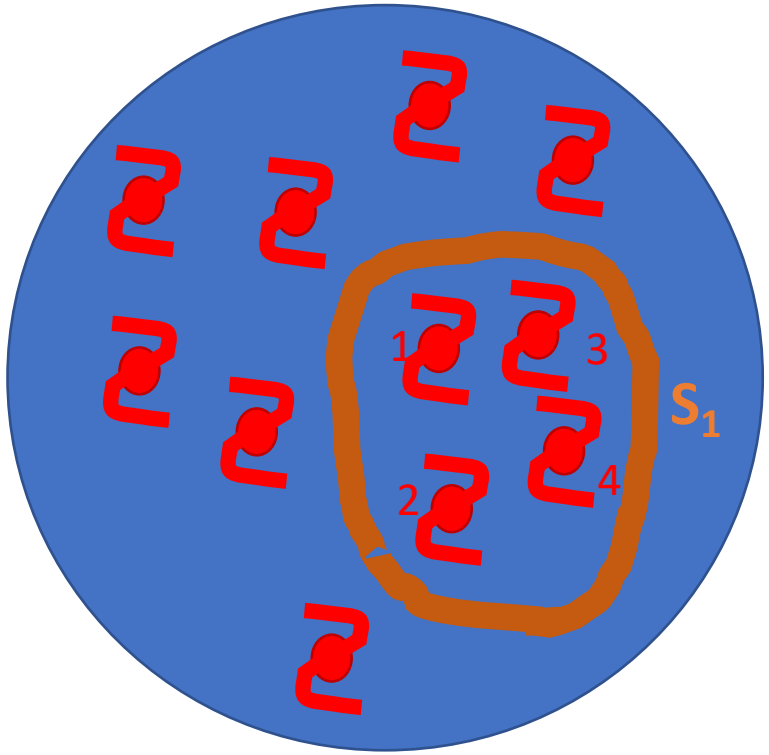
$S_1 = \{x_1, x_2, x_3, x_4\}$

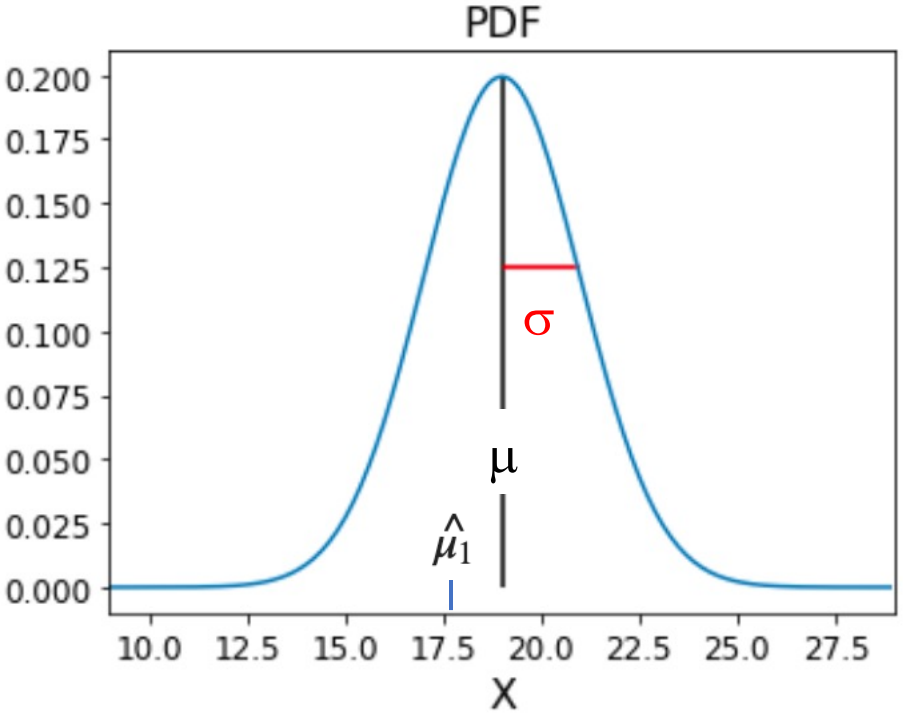
$$S_1 = \{19.45, 16.20, 16.43, 19.10\}$$

$$\hat{\mu}_1 = 17.8$$

Confidence interval from an estimator

If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$





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$$S_1 = \{19.45, 16.20, 16.43, 19.10\}$$

$$\hat{\mu}_1 = 17.8$$

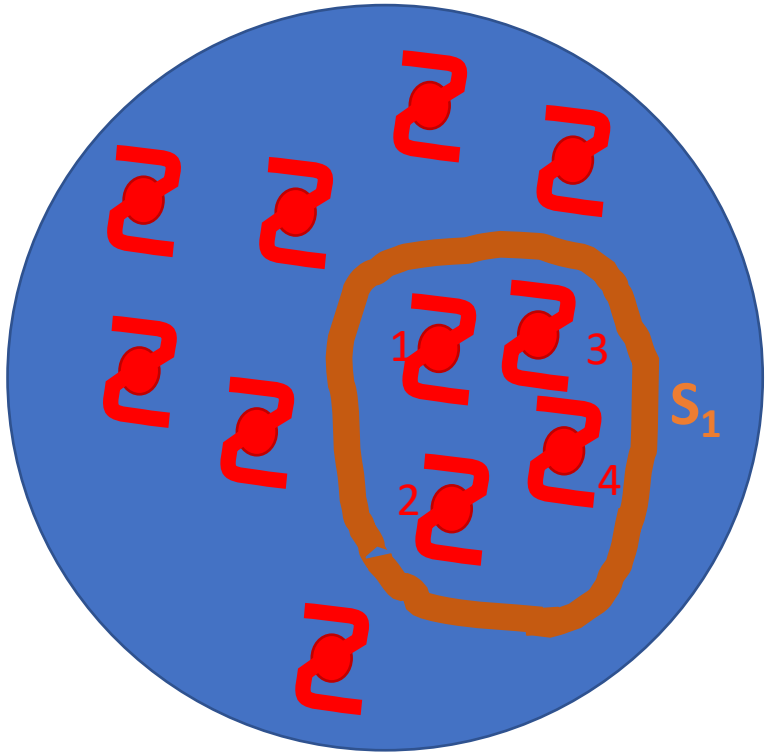
95.5 % CI on sample mean?

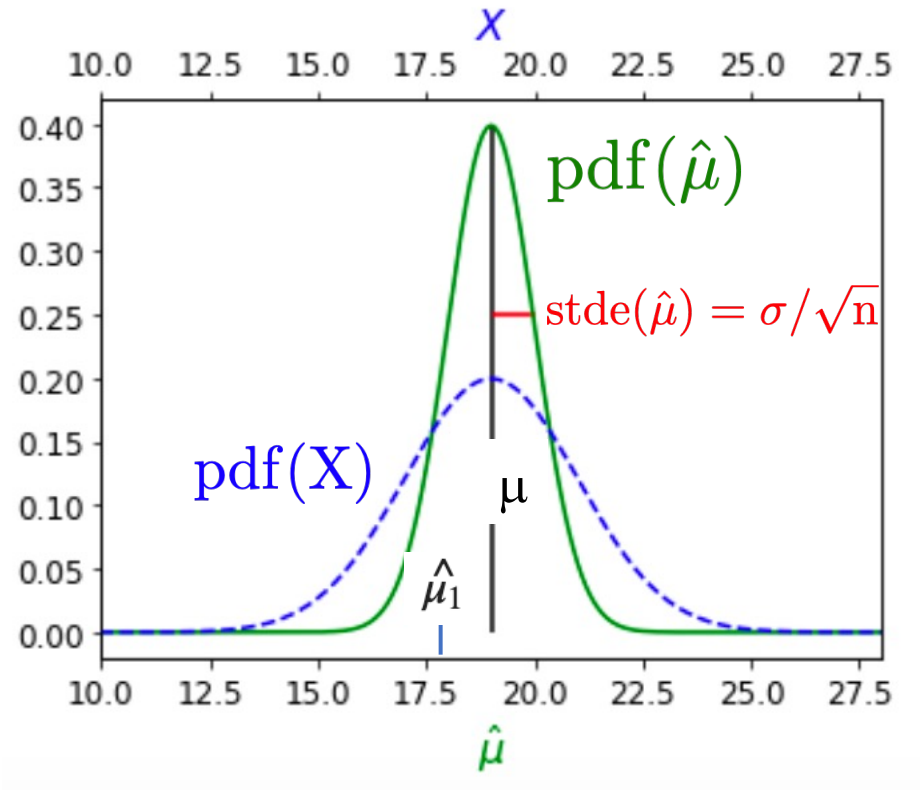
We need to look at the distribution of the sample mean

Confidence interval on the mean

95.5 % CI on $\hat{\mu}$?

If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$





Case 1: σ is known

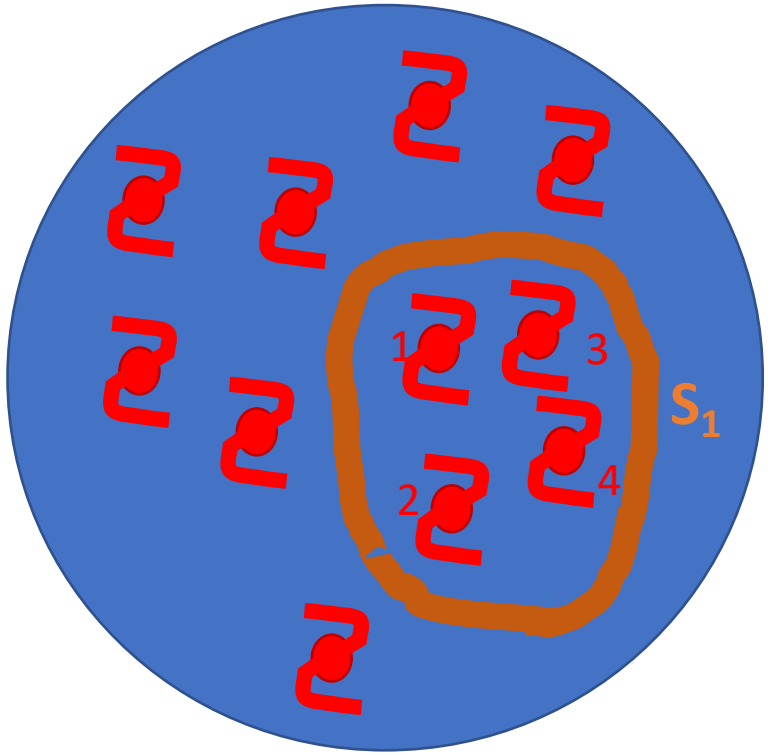
$$\boxed{\hat{\mu} \sim N(\mu, \sigma/\sqrt{n})}$$

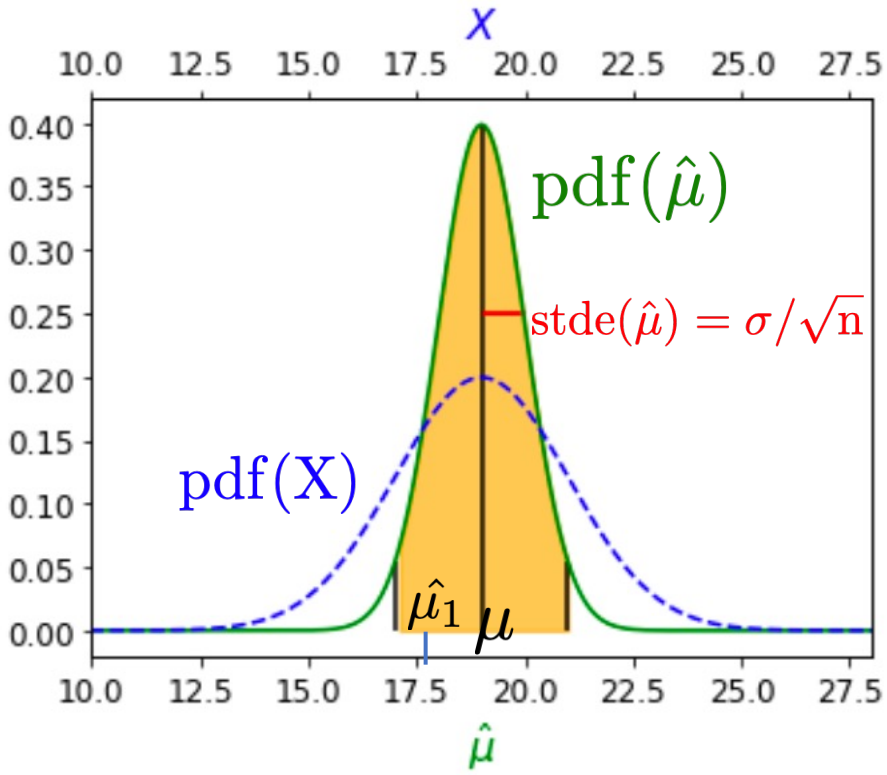
$$\mathrm{stde}(\hat{\mu}) = \sigma/\sqrt{\mathrm{n}}$$

Confidence interval on the mean

95.5 % CI on $\hat{\mu}$?

If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$





Case 1: σ is known

$$\boxed{\hat{\mu} \sim N(\mu, \sigma/\sqrt{n})}$$

$$\mathrm{stde}(\hat{\mu}) = \sigma/\sqrt{\mathrm{n}}$$

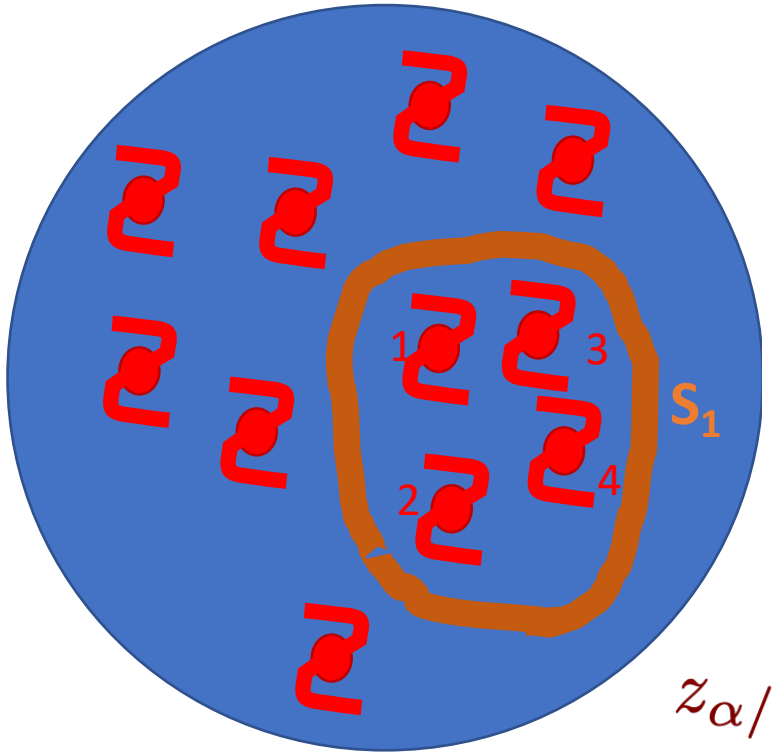
$$p(\hat{\mu}_1 \in [\mu \pm 2\,\mathrm{stde}]) = 0.955$$

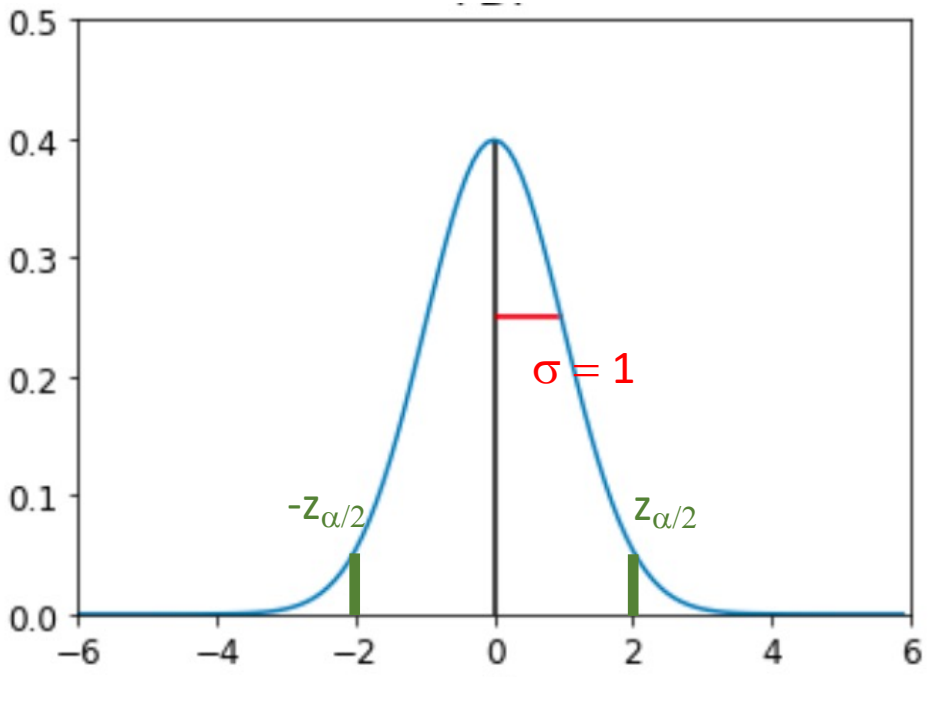
$$p(\mu \in [\hat{\mu}_1 \pm 2\,\mathrm{stde}]) = 0.955$$

Confidence interval on the mean

95.5 % CI on μ ?

If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$





$$z_{\alpha/2} = CDF^{-1}(1 - \alpha/2)$$

Case 1: σ is known

$$\mathrm{stde}(\hat{\mu}) = \sigma/\sqrt{n}$$

$$Z = \frac{\mu - \hat{\mu}}{\sigma/\sqrt{n}} \sim N(0,1)$$

p $(-z_{\alpha/2} < Z < z_{\alpha/2}) = \alpha * 100\ \%$

p $(-z_{\alpha/2} * \mathrm{stde} < \mu - \hat{\mu} < z_{\alpha/2} * \mathrm{stde})$

$= \alpha * 100\ \%$

$$p(\hat{\mu} - z_{\alpha/2}\,\mathrm{stde} < \mu < \hat{\mu} + z_{\alpha/2}\,\mathrm{stde}) = \alpha \times 100\%$$

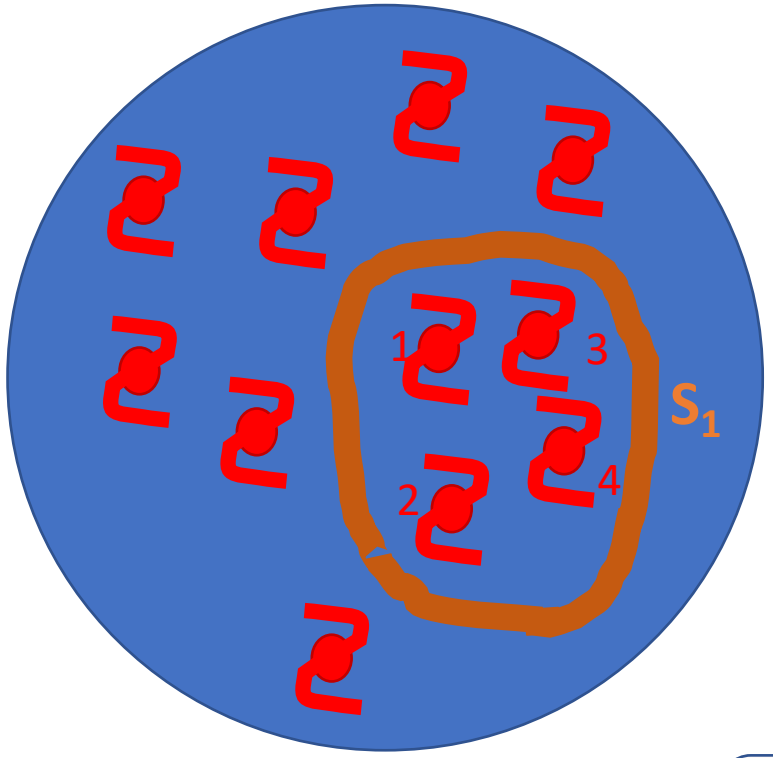
$$\mathrm{CI} \equiv [\hat{\mu} \pm \mathrm{m} \times \mathrm{stde}]$$

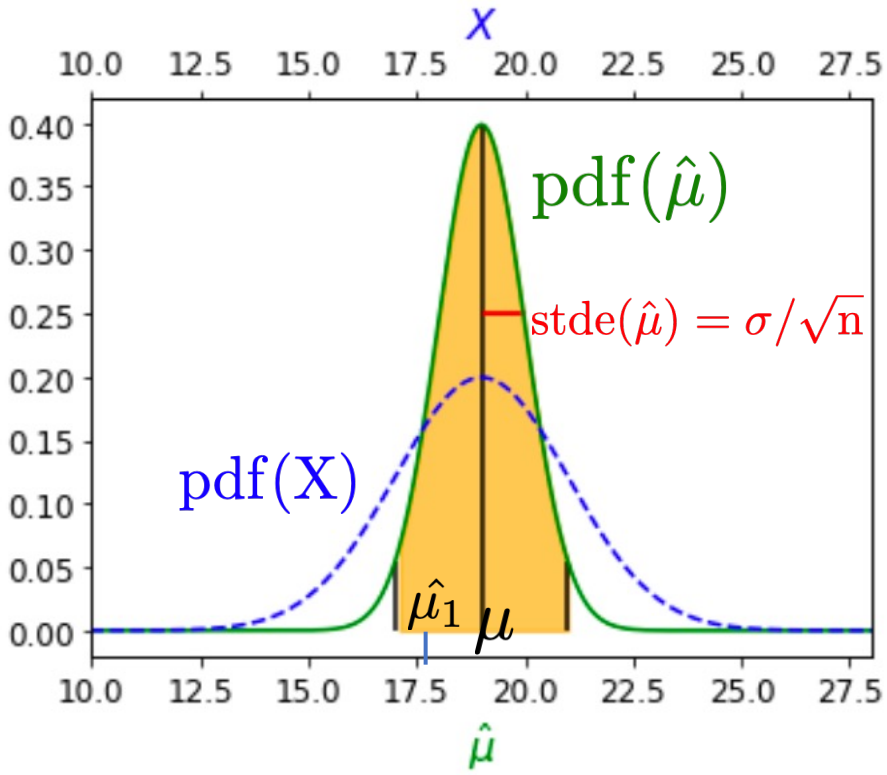
For 95.5 % CI, $z_{\alpha/2}$ = m = 2

Confidence interval on the mean

95.5 % CI on μ ?

If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$





Case 1: σ is known

$$\mathrm{stde}(\hat{\mu}) = \sigma/\sqrt{\mathrm{n}}$$

$$Z = \frac{\mu - \hat{\mu}}{\sigma/\sqrt{\mathrm{n}}} \sim N(0,1)$$

Case 2: σ is **un**known

$$\mathrm{stde}(\hat{\mu}) = \mathrm{s}/\sqrt{\mathrm{n}}$$

$$t = \frac{\mu - \hat{\mu}}{\mathrm{s}/\sqrt{\mathrm{n}}} \sim t(n-1)$$

$$\mathrm{CI} \equiv [\hat{\mu} \pm \mathrm{m} \times \mathrm{stde}]$$

Go to: Student_vs_Gauss.ipynb

Confidence interval

“Generic” strategy:

- Look at the PDF of a normalised/scaled estimator of known distribution.

- Define region around your (normalised) estimator (of know the distribution) that encloses **(1 - α) X 100 %** of the area under the PDF.

- For CI around the mean:

$$\mathrm{CI}_\alpha = [\mu - q_{\alpha/2}\,\mathrm{stde}, \mu + q_{\alpha/2}\,\mathrm{stde}]$$

$$q_{\alpha/2} = CDF^{-1}(1-\alpha/2)$$

For CI on other statistics (e.g. variance, difference between 2 means, proportions) : see Frequentist_inference_03.ipynb
(Material in Frequentist_inference_03.ipynb is supplementary material)

Go to: Sect. II.2. of the notebooks